Food and Agriculture Organization of the United Nations

Promoting productive water use and efficient water management in paddy fields

Piloting irrigation water monitoring system for paddy fields in Sri Lanka

THE PROJECT

In view of the projected global water demand, improved water use efficiency in irrigation is crucial to sustainably increase agricultural productivity. Paddy field systems are particularly water demanding, though products such as rice are not only a staple food but also constitute a major social and economic activity that provides public goods, as well as employment and income, to the rural population.

Funded by the Government of Japan, the project "Promoting productive water use and efficient water management in paddy fields" is implemented by the Land and Water Division of the Food and Agriculture Organization of the United Nations (FAO) in collaboration with partners and national institutions in Sri Lanka and Zambia. The project primarily consists of establishing pilot sites to demonstrate enhanced water management and capacity building of stakeholders at different levels on the multifunctional and value-added paddy fields in the two countries.

The pilot sites are equipped with flow-measuring devices to monitor and collect data on paddy water management and compare the performance of conventional and improved water management measures in paddy fields.

Output (1) of the project – the technical assessment of water management and demonstration of enhanced water management in paddy fields at pilot sites – aims to find evidence-based results on the benefits of enhanced water management through the following activities:

- selecting pilot demonstration sites and setting up an irrigation water monitoring system for paddy fields;
- monitoring and collecting irrigation data at the pilot site; and
- implementing pilot demonstration sites to compare conventional and enhanced water management measures in paddy fields.

COUNTRY PROFILE

Rice is the staple food in Sri Lanka, with the average rice consumption at about 90 kg per capita per year. Some 8.1 million families or 72 percent of rural households cultivate rice for their livelihoods, making rice production the largest contributor to food security and rural economy. Over 50 percent of the rice production comes from farms with less than 0.4 ha production area, providing jobs and food to the smallholders.

Rice is cultivated in two distinct cropping seasons throughout the country. The main cropping season is Maha, which lasts from late September to late March and is fed by the second intermonsoon and the northeast monsoon rains. A subsequent rice crop is grown in the second growing season called Yala. It lasts from early April to early September and is fed by the first intermonsoon rains and the southwest monsoon, which mainly brings rain to the southwest of the country.

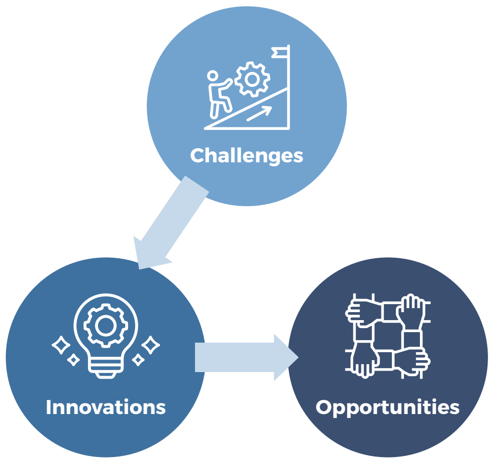
Sri Lanka is highly vulnerable to climate change and droughts due to its heavy reliance on monsoon rainfall for crop production, particularly during the Yala season. In 2016–2017, rice production was severely affected when the country faced its worst drought in 40 years. With rainfall levels drastically below average, rice production decreased by nearly 40 percent. This sharp decline in staple crop production disrupted national food security and rural livelihoods, causing an economic fallout. The drought underscored the urgent need for adaptive water management strategies to mitigate climate-related risks in agriculture.

TURNING CHALLENGES INTO OPPORTUNITIES

Traditional rice farming in Sri Lanka consumes nearly 80 percent of the nation's freshwater resources that are already highly vulnerable to drought. This water-intensive practice not only results in significant non-beneficial water use but also creates an anaerobic environment in flooded fields, leading to considerable greenhouse gas emissions. Addressing these challenges presents opportunities for environmental sustainability and socioeconomic benefits.

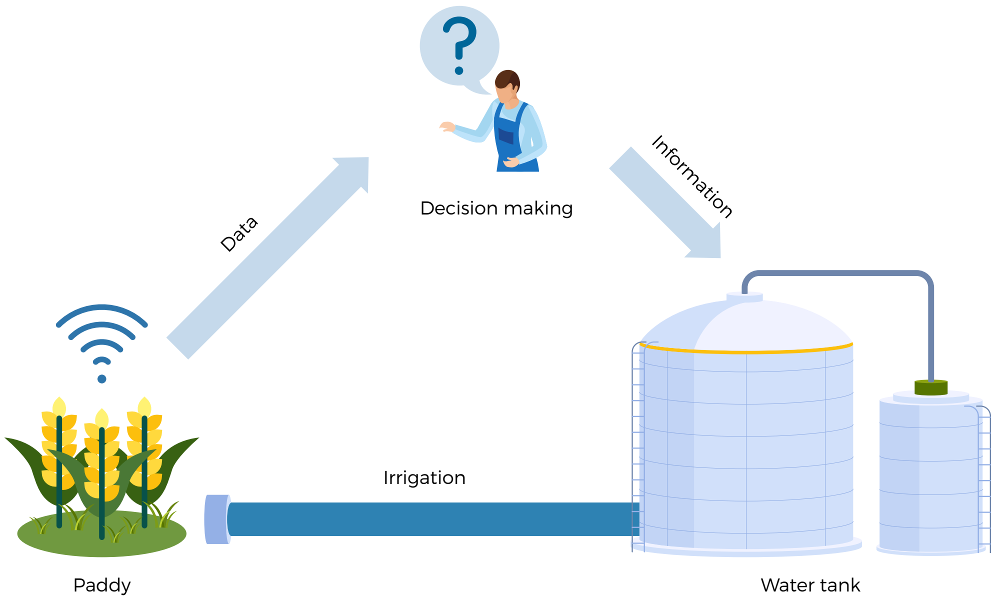
Optimizing water use can enhance productivity, strengthen drought resilience, and benefit communities and ecosystems. To tackle these issues, the project recognized a suite of advanced farming practices, including an adaptive irrigation technique called alternate wetting–drying (AWD). This method allows rice paddies to dry down between flood irrigations after the initial crop development, reducing water usage and fostering an aerobic root zone that helps mitigate greenhouse gas emissions while improving resource use efficiency and farm productivity.

FIGURE 1. From challenges to opportunities in rice farming water management



Source: Authors' own elaboration

FIGURE 2. **Schematic of irrigation monitoring system using 3G/4G GSM communication for data transmission**



Source: Authors' own elaboration

THE NEED FOR AN IRRIGATION WATER MONITORING SYSTEM

The development of good water management practices in paddies requires empirical evidence. Water and environment-related data are necessary to understand causal relationships and define strategies for improvement. In particular, AWD in large fields, though more effective than in small fields, could be challenging. It requires monitoring the precise water level during the wetting cycle and soil moisture in the root zone during the drying cycle to optimize water use and crop yield. An accurate estimate of each irrigation dose is important to withdraw the allocated share of water from the storage tank and to estimate water productivity and water use efficiency. Manual data collection of all these parameters is labor-intensive, time-consuming and prone to inaccuracies.

To overcome these limitations and ensure precise monitoring, a comprehensive irrigation monitoring system was essential to track flow discharge, water levels and soil moisture in near real-time, facilitating data-driven decision-making for efficient water management in paddies.

Site selection for the irrigation water monitoring system

The pilot site was selected based on certain technical criteria. The experimental and control fields represent over 50 percent of smallholder rice farms in Sri Lanka. Both fields are irrigated from a small storage tank with a surface area of about half a hectare and a catchment area of approximately one square kilometer. The tank water is shared by 25 farmers.

Due to the location of the site in the intermediate climatic zone prone to water scarcity, farmers rely on supplemental irrigation during the Yala season, making it ideal for testing varying water management practices, including AWD. The feasibility of the irrigation infrastructure of the site was a key criterion in its selection. The gated outlet of the storage tank and separate water intakes for the experimental and control plots enabled the installation of independent monitoring systems in both plots with regulated water flow. Accessibility via an inspection road was also prioritized to ensure year-round access for system maintenance. Having both fields owned and managed by the same farmer allowed for consistency and reliability, as differences in outcomes could be attributed solely to the experimental intervention and not to the variation in farming skills or external conditions.

The irrigation water monitoring system and its components

Assembling an appropriate irrigation monitoring system was critical to ensure appropriate and precise measurements with economies of scale. A modular design approach was adopted, which means individual data loggers with relevant sensors were installed separately at the experimental and control plots. Hydrostatic-type pressure sensors with a measurement range of 0–100 cm were installed to monitor the water level in a flume fitted with a V-notch for flow discharge measurement. Due to their robust design and high accuracy (0.5 percent), the same sensors were used to measure standing water levels in the paddy fields. An AWD perforated tube was installed in each field, extending 5 cm above the ground surface and 17 cm below it, with the sensor positioned 10 cm below the ground.

This setup allowed for water level measurements both above the ground surface during wetting cycles and below the surface during drying cycles.

The adaptability of the monitoring system to diverse physical and environmental conditions of paddy fields was a concern so the sensors and data loggers were chosen to comply with IP-68 rating to sustain moisture ingress in harsh paddy environments.

As a backup, capacitive-type soil moisture sensors with a measurement range of 0–100 percent relative to field capacity were installed at depths of 10 cm and 30 cm to alert for excessive moisture depletion in case of water level sensor malfunction. In a modular setting, all the sensors from each plot were connected to a measurement and control data logger capable of operating on a 4G GSM network for real-time data transmission.

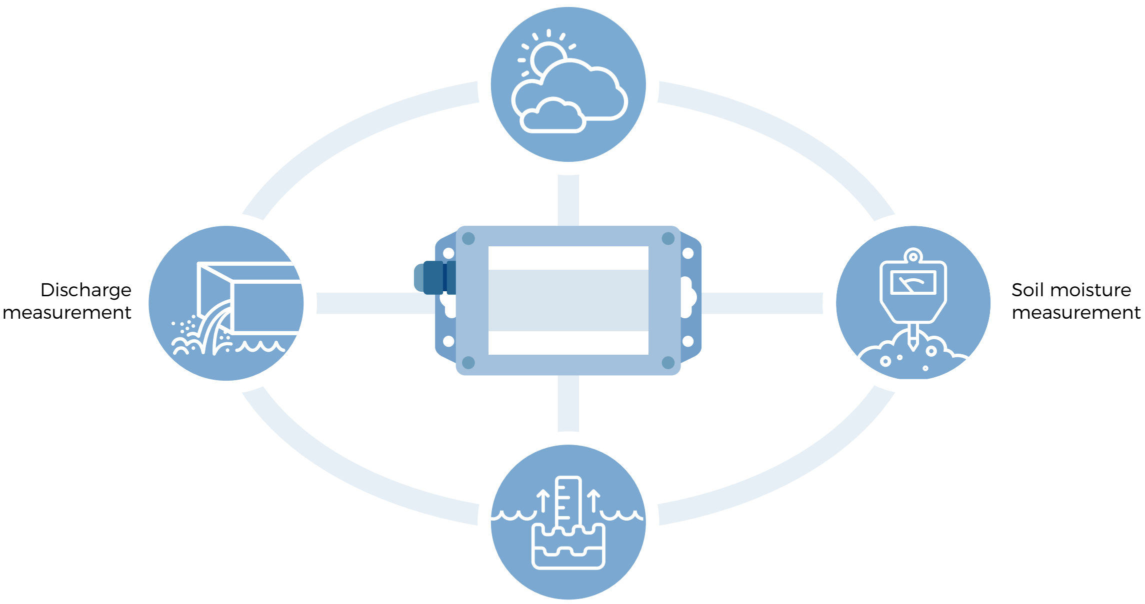
The data loggers are powered by solar-rechargeable batteries to ensure year-round operation. Data from the two loggers at 30-minute measurement and 6-hour transmission frequency were routed to an Internet-of-Things cloud platform accessible via a user-friendly web interface. The system enabled near-real-time monitoring of irrigation frequency, water volumes, paddy water levels and soil moisture at two depths.

TABLE 1. Technical specification of the system components

Data logger	Water level sensor	Soil moisture sensor
Type: Measurement and control	Type: Hydrostatic	Type: Capacitive
Connectivity: Global 4G	Measurement range: 0–100 cm	Measurement range: 0–100 percent
Reading ports: 6 No	Enclosure: Stainless steel	Enclosure: HD plastic
Memory type: EPROM	Rating: IP68 (1.5 m depth)	Rating: Weather- and waterproof
Digital precision: 0.01 percent (16 bit)	Cable: Vented (correcting atm)	Connection: 3 wire cables
Rating: IP-68 (1.5 m depth)	Accuracy: 0.5 percent	Accuracy: High
Configuration: Remotely through 4G	Compliance: ISO 9001:2015	Temp: -20 – 80 °C
Operation: Battery, Solar		

Source: Authors' own elaboration

FIGURE 3. Stylized figure of the monitoring system components



Source: Authors' own elaboration

Daily weather data, including rainfall, was recorded at the nearby Rice Research and Development Institute using an automated weather station. The effectiveness of an irrigation water monitoring system lies in its ability to provide real-time information for timely decision-making on the amount and timing of subsequent irrigations. The system provides alerts and notifications to decision-makers when water levels deviate from the desired range, helping prevent overirrigation or crop stress.

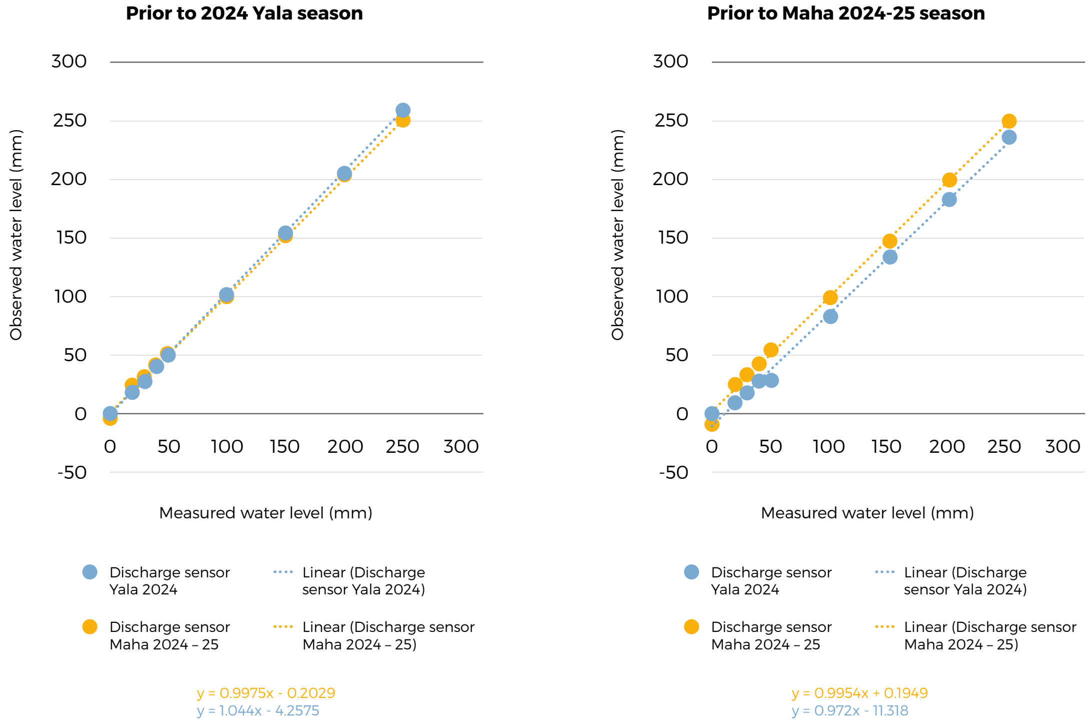
The selected monitoring system is cost-effective in terms of initial investment and operation and maintenance cost as it relies on off-the-shelf technology and existing mobile communication medium.

Evaluation of results

Before the start of each cropping season, the monitoring system and the sensors underwent a comprehensive inspection, followed by calibration and trial runs in a controlled laboratory environment to ensure accurate measurements. This process is particularly critical for hydrostatic sensors that are susceptible to lagging or leading sensor drift caused by factors such as mechanical wear, aging components, water quality variations, and residue buildup during the previous seasonal installations.

The effect of leading drift is more pronounced in paddy water level sensors that remain submerged in standing water and are exposed to high sediment levels, fertilizers and crop nutrients throughout the season, as compared to discharge sensors that typically operate at the intake in comparatively cleaner tank water. If left uncorrected, this drift can lead to considerable inaccuracies in data collection. To address this, calibration constants derived from linear regression equations are updated seasonally through the online interface of data loggers, ensuring continuous accuracy and reliability of the system.

FIGURE 4. Calibration of hydrostatic level sensors prior to the two cropping seasons



Source: Authors' own elaboration

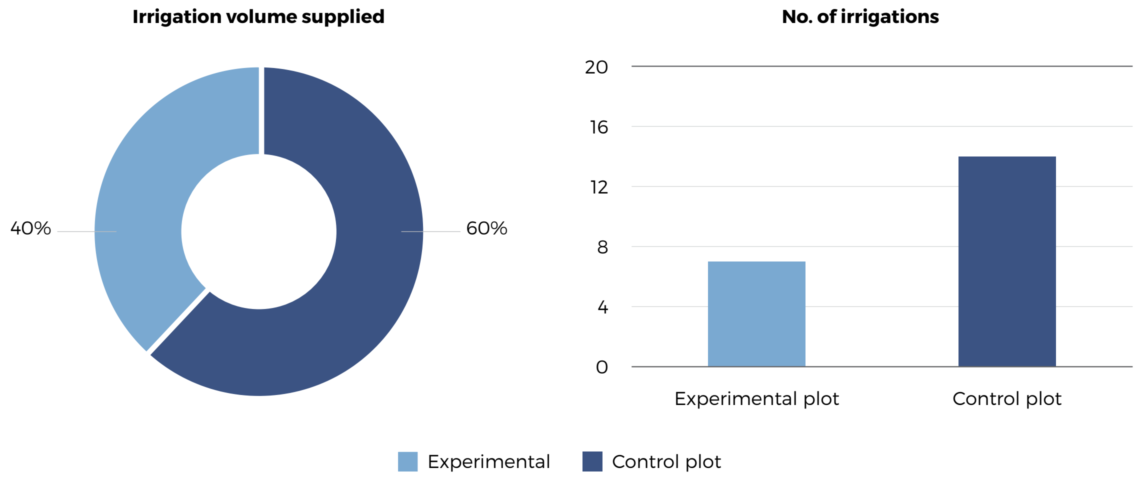
© FAO/Priyadharshanee Herath

During the pilot phase, the monitoring system demonstrated its effectiveness in enhancing irrigation water management through smart decision-making. By integrating the suite of productivity and efficiency measures, including AWD, the system was also able to optimize the use of both rainfall and irrigation water at the experimental plot. This led to a combined reduction of 40 percent in water application compared to the control plot during the Yala 2024 season.

Real-time data from the system across multiple dimensions (e.g. weather, paddy water level and soil moisture) enabled informed decision-making for irrigation scheduling. As a result, the number of irrigation events in the experimental plot was reduced by 50 percent, from 14 in the control plot down to 7 in the experimental plot. Moreover, the overall normalized depth of irrigation water applied during the season was significantly lower: 23 cm for the experimental plot versus 42 cm for the control plot. This decision to practice AWD allowed the use of around 25 percent less water in the experimental plot compared to the control plot during the Yala 2024 season.

Despite the reduced water volume, the pilot plot yielded higher farm productivity, profitability and crop water productivity. These results underscore how leveraging advanced monitoring and adaptive water management strategies not only conserves water but also maintains the productivity of the experimental plot. The monitoring system proved smart during the pilot phase. The decision-making in implementing the suite of productivity and efficiency measures, including AWD, made effective use of rainfall and irrigation water at the experimental plot.

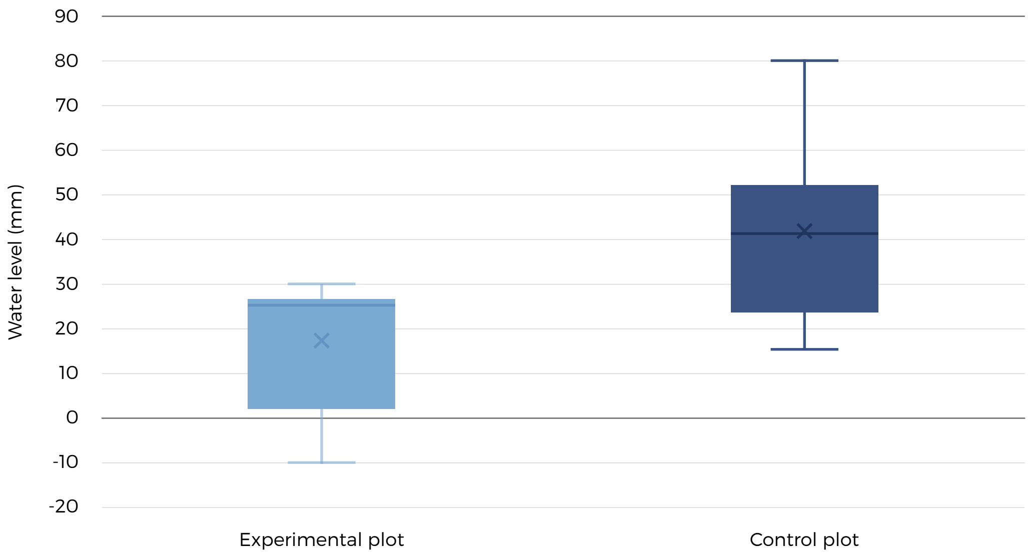
FIGURE 5. Percent volume and number of irrigations applied at the two plots



Source: Authors' own elaboration

The effective performance of the irrigation water monitoring system is also demonstrated by its ability to precisely track the paddy water levels. The submerged installation of the water level sensor inside the AWD tube at the experimental plot proved useful in capturing the drying cycles, which is critical in determining the start of the next irrigation under the AWD practice. The system recorded a mean water level of 17 mm in the experimental plot, with fluctuations ranging from 30 mm to -10 mm. This close-to-zero water level range indicates effective water management and controlled drying cycles. In contrast, the control plot, which used the traditional continuous flooding method, exhibited a higher average water level of 42 mm with a broader fluctuation range between 80 mm and 15 mm. The wider range in the control plot reflects less precise water management, highlighting the advantage of using the monitoring system in the AWD approach.

FIGURE 6. Paddy water level monitoring at the two plots



Source: Authors' own elaboration

The scalability strategy

The irrigation water monitoring system offers transformative benefits in the context of sustainable water use, farm productivity and profitability. The localized control and precise measurement offered by individual monitoring units allow farmers to manage their irrigation practices under changing environmental conditions. Each unit can be calibrated to a particular crop's unique environmental conditions and water needs, enabling a more refined and responsive approach to water management.

With real-time data streaming from such individual units, farmers can dynamically adjust irrigation schedules to match the actual needs of their crops. This means that water is applied only when necessary, reducing waste and ensuring that crops receive the optimal amount of irrigation for crop growth. The efficient use of water not only supports better crop yields but also translates into significant cost savings and environmental benefits.

Moreover, the data-driven information provided by the system allows for continuous refinement of irrigation strategies. Over time, this leads to a more sustainable and resilient agricultural practice as farmers are better equipped to adapt to seasonal variations and changing environmental conditions. Precise control over irrigation application also helps minimize environmental issues such as nutrient runoff and soil erosion, contributing to the broader goals of sustainability. The modular nature of the system means that it is inherently scalable and flexible. Farmer communities can easily integrate additional sensors or upgrade existing units as they expand smart irrigation, ensuring that the system remains effective even as farm sizes and environmental conditions change. This flexibility not only enhances operational resilience but also supports long-term planning and investment, leading to sustained improvements in productivity and profitability.

BOX 1. CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

The irrigation water monitoring system contributes significantly to several Sustainable Development Goals (SDGs). For SDG 1, which focuses on ending poverty, the system can optimize water resources management, which translates to reduced irrigation costs and improved farm productivity. By enabling farmers to use water more efficiently and reducing the risk of crop failure, this technology can help increase incomes and economic resilience in rural communities such as in Kurunigala, the pilot area in Sri Lanka. As farmers achieve higher yields with lower resource inputs, the financial stability of the communities improves, contributing to poverty reduction.

In addressing SDG 2, Zero Hunger, the system plays a pivotal role by ensuring that crops receive the optimal amount of water at the right time. The ability to monitor water levels accurately allows for the implementation of precision irrigation techniques, such as alternate wetting–drying (AWD). This leads to better crop growth and consistent yields, directly contributing to food security and sustainable agricultural practices. By maximizing resource efficiency and reducing waste, the system supports a more reliable food supply, which is essential in the broader fight against hunger and malnutrition.

The impact of the monitoring system on water management aligns closely with SDG 6, which is dedicated to clean water and sanitation. By carefully controlling water applications, the system not only helps conserve water but also helps maintain its quality by minimizing runoff and deep percolation from rice paddies. This sustainable management of water resources ensures that clean water is available for agricultural and domestic needs, thereby enhancing environmental conservation efforts.

© FAO/Waqas Ahmad

LESSONS LEARNED

Irrigation water monitoring systems equip farmers with the tools to implement precision irrigation practices that conserve water, optimize resource use, and ultimately drive higher crop yields and economic returns. This holistic approach to water management is critical in moving towards a more sustainable and profitable agricultural future. The monitoring system was an integral part of the suite of productivity and efficiency measures introduced in this project. In this context, several valuable messages were concluded from the pilot:

- The pilot demonstrated that adaptive water management strategies are crucial in dealing with water scarcity, particularly in areas that have large populations and limited water availability, such as the intermediate climatic zone of Sri Lanka. By adopting good water management practices, including the AWD technique, the pilot demonstrated substantial water saving while optimizing crop yield. This approach not only conserved water but also maintained farm productivity and income, highlighting the importance of flexibility and responsiveness in irrigation practices to mitigate the risks posed by climate change and drought.

- The pilot installation represented over 50 percent of smallholder rice farms in Sri Lanka, showcasing its relevance to the wider farming community. The positive results achieved in the experimental plot – where water levels were closely monitored and irrigation events reduced by 50 percent –

demonstrate that such a system can be scaled up to benefit small-scale farmers across the country. This adaptability is significant for food security and the rural economy of Sri Lanka, where rice cultivation supports millions of livelihoods.

- The modular design of the monitoring system facilitated its successful installation and operation. Each data logger unit with a set of sensors was independently installed, enabling a flexible setup suited to the specific requirements and environment of the farming plots. This design approach not only simplified the installation process but also ensured that the system could be easily expanded or upgraded as needed, supporting its potential for widespread adoption and scalability in various farming contexts.

- The availability of near-real-time data on water levels, soil moisture and irrigation schedules proved to be a gamechanger for smart water management at farm scale. The capability of the system to precisely capture critical data allowed farmers to make informed decisions about irrigation timing and volumes, reducing waste and preventing crop stress. This data-driven approach ensures more efficient use of water resources, contributes to better crop management, and fosters resilience against climatic uncertainties.

- By accurately tracking and managing water flows, the monitoring system promoted a more equitable distribution of water among the 25 farmers sharing the same storage tank. This equitable access to water resources helped prevent disputes and ensured that all farmers benefit equally from the shared water resource.

- The use of off-the-shelf technology and integration with existing mobile communication networks made the monitoring system cost-effective both in terms of initial investment and operational and maintenance cost. The reliance on solar-rechargeable batteries enhanced its sustainability, making it suitable for remote agricultural areas with limited grid-power infrastructure.

- The success of the monitoring system in reducing irrigation requirements while maintaining productivity underscores its role in building resilience against climate variability. By enabling more precise and timely irrigation, farmers can better navigate fluctuations in water availability and adapt to changing environmental conditions.

For more information, please contact:
Eva Pek, Land and Water Officer, Land and Water Division
Komei Sasa, Junior Professional Officer, Land and Water Division
Priyadharshanee Herath, Natural Resources Management Expert, FAO-Sri Lanka.
Waqas Ahmad, Water Resources Management Specialist, Land and Water Division
Maher Salman, Senior Land and Water Officer/ AWM Team Lead, Land and Water Division

With the financial support of the Ministry of Agriculture, Forestry and Fisheries, Japan

With the technical support of the Rice Research and Development Institute